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# Formatting Instructions For NeurIPS 2025

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## Abstract

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## 2 1 Introduction

3 In the high stakes world of algorithmic decision making, the outcomes of machine learning models  
4 can have significant impacts on individuals and society in areas such as healthcare, criminal justice,  
5 lending, hiring. In this contexts, fairness, is not just a desirable property but a critical requirement  
6 TODO: [ADD CITATIONS]. In particular, group fairness, which requires predictive behaviour  
7 remains equitable across demographically defined groups, has emerged as a key focus of research in  
8 this area.

9 Notwithstanding the significant progress in developing group fairness algorithms, most of the existing  
10 work assumes a stationary data distribution which, in the real world, rarely holds. Instead, data  
11 distributions evolve over time: policy shifts, seasonal effects and changes in societal norms can lead  
12 to distribution shifts (often referred to as concept drift) that can undermine predictive performance  
13 and fairness guarantees.

14 While online learning algorithms have been created to preserve predictive accuracy, there has been  
15 less work on how to preserve fairness under concept drift. This is a critical gap, because fairness  
16 violations may remain undetected longer than accuracy degradation, disrupting the trustworthiness of  
17 machine learning systems and leading to significant harms.

18 To address this challenge it is important not only to detect distributional changes as they occur but, at  
19 the same time, characterize their impact on fairness and react accordingly. In this paper, we study  
20 fairness-aware learning under concept drift in online settings and investigate how a novel framework  
21 for detecting and responding to concept drift can be used to preserve fairness guarantees over time.

## 22 2 Experiments

23 Tables 1–5 report per-scenario results on Adult with mean  $\pm$  standard deviation across seeds from  
24 files/experiments. Aranyani results use 4 available seeds; ARF and RFR use 5 seeds.

Table 1: No drift scenario results on Adult (mean  $\pm$  std over seeds)

| Model    | Accuracy $\uparrow$                   | DP $\downarrow$                       |
|----------|---------------------------------------|---------------------------------------|
| Aranyani | <b>0.8538 <math>\pm</math> 0.0041</b> | 0.0860 $\pm$ 0.0026                   |
| ARF      | 0.7700 $\pm$ 0.0000                   | <b>0.0336 <math>\pm</math> 0.0000</b> |
| RFR      | 0.8050 $\pm$ 0.0079                   | 0.0585 $\pm$ 0.0073                   |

Table 2: Abrupt gender scenario results on Adult (mean  $\pm$  std over seeds)

| Model    | Accuracy $\uparrow$                   | DP $\downarrow$                       |
|----------|---------------------------------------|---------------------------------------|
| Aranyani | <b>0.8607 <math>\pm</math> 0.0199</b> | 0.0414 $\pm$ 0.0192                   |
| ARF      | 0.7700 $\pm$ 0.0000                   | <b>0.0336 <math>\pm</math> 0.0000</b> |
| RFR      | 0.8030 $\pm$ 0.0045                   | 0.0622 $\pm$ 0.0042                   |

Table 3: Gradual gender scenario results on Adult (mean  $\pm$  std over seeds)

| Model    | Accuracy $\uparrow$                   | DP $\downarrow$                       |
|----------|---------------------------------------|---------------------------------------|
| Aranyani | <b>0.8573 <math>\pm</math> 0.0026</b> | <b>0.0129 <math>\pm</math> 0.0073</b> |
| ARF      | 0.7700 $\pm$ 0.0000                   | 0.0336 $\pm$ 0.0000                   |
| RFR      | 0.8040 $\pm$ 0.0074                   | 0.0690 $\pm$ 0.0118                   |

## References

- References follow the acknowledgments in the camera-ready paper. Use unnumbered first-level heading for the references. Any choice of citation style is acceptable as long as you are consistent. It is permissible to reduce the font size to small (9 point) when listing the references. Note that the Reference section does not count towards the page limit.
- [1] Alexander, J.A. & Mozer, M.C. (1995) Template-based algorithms for connectionist rule extraction. In G. Tesauero, D.S. Touretzky and T.K. Leen (eds.), *Advances in Neural Information Processing Systems 7*, pp. 609–616. Cambridge, MA: MIT Press.
- [2] Bower, J.M. & Beeman, D. (1995) *The Book of GENESIS: Exploring Realistic Neural Models with the GEneral NEural Simulation System*. New York: TELOS/Springer-Verlag.
- [3] Hasselmo, M.E., Schnell, E. & Barkai, E. (1995) Dynamics of learning and recall at excitatory recurrent synapses and cholinergic modulation in rat hippocampal region CA3. *Journal of Neuroscience* **15**(7):5249-5262.

## A Technical Appendices and Supplementary Material

Technical appendices with additional results, figures, graphs and proofs may be submitted with the paper submission before the full submission deadline (see above), or as a separate PDF in the ZIP file below before the supplementary material deadline. There is no page limit for the technical appendices.

Table 4: Gender-relationship decouple scenario results on Adult (mean  $\pm$  std over seeds)

| Model    | Accuracy $\uparrow$                   | DP $\downarrow$                       |
|----------|---------------------------------------|---------------------------------------|
| Aranyani | <b>0.8516 <math>\pm</math> 0.0052</b> | 0.0854 $\pm$ 0.0065                   |
| ARF      | 0.7700 $\pm$ 0.0000                   | <b>0.0336 <math>\pm</math> 0.0000</b> |
| RFR      | 0.7890 $\pm$ 0.0096                   | 0.0713 $\pm$ 0.0142                   |

Table 5: Occupation gender reversal scenario results on Adult (mean  $\pm$  std over seeds)

| Model    | Accuracy $\uparrow$                   | DP $\downarrow$                       |
|----------|---------------------------------------|---------------------------------------|
| Aranyani | <b>0.8516 <math>\pm</math> 0.0056</b> | 0.0524 $\pm$ 0.0060                   |
| ARF      | 0.7700 $\pm$ 0.0000                   | <b>0.0336 <math>\pm</math> 0.0000</b> |
| RFR      | 0.7990 $\pm$ 0.0124                   | 0.0728 $\pm$ 0.0057                   |

## 42 NeurIPS Paper Checklist

43 The checklist is designed to encourage best practices for responsible machine learning research,  
 44 addressing issues of reproducibility, transparency, research ethics, and societal impact. Do not remove  
 45 the checklist: **The papers not including the checklist will be desk rejected.** The checklist should  
 46 follow the references and follow the (optional) supplemental material. The checklist does NOT count  
 47 towards the page limit.

48 Please read the checklist guidelines carefully for information on how to answer these questions. For  
 49 each question in the checklist:

- 50 • You should answer [Yes], [No], or [NA].
- 51 • [NA] means either that the question is Not Applicable for that particular paper or the  
 52 relevant information is Not Available.
- 53 • Please provide a short (1–2 sentence) justification right after your answer (even for NA).

54 **The checklist answers are an integral part of your paper submission.** They are visible to the  
 55 reviewers, area chairs, senior area chairs, and ethics reviewers. You will be asked to also include it  
 56 (after eventual revisions) with the final version of your paper, and its final version will be published  
 57 with the paper.

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 60 proper justification is given (e.g., "error bars are not reported because it would be too computationally  
 61 expensive" or "we were unable to find the license for the dataset we used"). In general, answering  
 62 "[No]" or "[NA]" is not grounds for rejection. While the questions are phrased in a binary way, we  
 63 acknowledge that the true answer is often more nuanced, so please just use your best judgment and  
 64 write a justification to elaborate. All supporting evidence can appear either in the main paper or the  
 65 supplemental material, provided in appendix. If you answer [Yes] to a question, in the justification  
 66 please point to the section(s) where related material for the question can be found.

67 IMPORTANT, please:

- 68 • Delete this instruction block, but keep the section heading "NeurIPS Paper Checklist",
- 69 • Keep the checklist subsection headings, questions/answers and guidelines below.
- 70 • Do not modify the questions and only use the provided macros for your answers.

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72 Question: Do the main claims made in the abstract and introduction accurately reflect the  
 73 paper's contributions and scope?

74 Answer: **[TODO]**

75 Justification: **[TODO]**

76 Guidelines:

- 77 • The answer NA means that the abstract and introduction do not include the claims  
 78 made in the paper.
- 79 • The abstract and/or introduction should clearly state the claims made, including the  
 80 contributions made in the paper and important assumptions and limitations. A No or  
 81 NA answer to this question will not be perceived well by the reviewers.
- 82 • The claims made should match theoretical and experimental results, and reflect how  
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- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

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